25900897 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCMBG101 - WONDERS OF THE MICROBIAL WORLD

(2024 ADMISION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Name the Father of Immunology [U] / [CO1]

   a). Robert Koch b). Edward Jenner

   c). Francesco Redi d). Louis Pasteur

2. In his discovery of antibiotics, which bacteria was affected by the antibiotic produced by Penicillium? [U] / [CO1]

   a). E. coli b). Salmonella

   c). Pseudomonas d). Staphylococcus

3. The protein that make up flagella is called ..... [K] / [CO3]

   a). pilin b). flagellin

   c). filament d). porin

4. Name the Father of Antiseptic Surgery [K] / [CO1]

   a). Paul Ehrlich b). Joseph Lister

   c). Robert Koch d). Alexander Fleming

5. Bacteria that is most widely employed in bioleaching [U] / [CO1]

a). Pseudomonas b). Clostridium

c). Thiobacillus d). Bacillus

6. The arrangement of bacteria in groups of 8 cells is called ..... [U] / [CO3]

a). Tetrads b). Streptococci

c). Diplococci d). Sarcinae

7. Select the media that is employed widely for assaying antimicrobial activity by disc diffusion method. [K] / [CO4]

a). Potato dextrose agar b). Mueller Hinton agar

c). Nutrient agar d). MacConkey agar

8. How does the capsule forming bacteria appear after Anthony staining method? [U] / [CO2]

a). Cell in violet colour, capsule is colourless b). Cell in red colour, capsule in violet colour

c). Cell in deep violet colour, capsule in light violet colour d). Capsule appear as colourless halos surrounding the stained bacterial cell

9. In fluorescence microscope, which of the following performs the function of removing all the long wavelength light from falling on the specimen? [U] / [CO2]

a). Exciter filter b). Barrier filter

c). Mercury arc lamp d). Dichroic mirror

10. Which antibiotic class is known for inhibiting folic acid synthesis in bacteria? [K] / [CO4]

a). Cephalosporins b). Tetracyclines

c). Sulfonamides d). Penicillins

[10x1 = 10]

Part B

Fill in the Blanks

Answer **All** Questions

Each Question carries **1** mark

11. Holding method of pasteurization uses a temperature of ..... and time of ..... [K] / [CO4]

12. ..... is known as the Father of Microbiology [K] / [CO1]

13. ..... is the primary stain used in Gram staining. [A] / [CO2]

14. The envelope of a virus is derived from the ..... of the host. [U] / [CO3]

15. ..... is the solidifying agent used in the preparation of solid media [U] / [CO1]

[5x1 = 5]

Part C

State whether True or False

Answer **All** Questions

Each Question carries 1 mark

16. Louis Pasteur proposed the theory of biogenesis. [K] / [CO1]

17. The cells have the same refractive index in all regions and light will pass through them uniformly [A] / [CO2]

18. Water acts as the decolourizing agent in endospore staining. [K] / [CO2]

19. Bacteriophage multiplication takes place by lytic and lysogenic cycle. [U] / [CO3]

20. HTST method uses a temperature of 63 degree Celcius for 15 seconds. [K] / [CO4]

[5x1 =5]

Part D

Very Short Answer Type Questions

Answer any **10** Questions

Each Question carries **1** mark

21. Who is referred to as the Father of Chemotherapy and why? [U] / [CO1]

22. Explain theory of abiogenesis [U] / [CO1]

23. Name two non-acid-fast bactera [K] / [CO2]

24. Name two basic stains [K] / [CO2]

25. How does a dichroic mirror contribute to image clarity in fluorescence microscopy? [U] / [CO2]

26. What is a smear? [U] / [CO2]

27. What are endospores? [K] / [CO3]

28. What do you mean by imperfect fungi? [K] / [CO3]

29. Draw the diagram of TMV. [K] / [CO3]

30. Explain the mode of action of chlorine on microorganisms. [U] / [CO4]

31. What are the standard time and temperature guidelines for autoclaving? [U] / [CO4]

32. What is the primary purpose of iris diaphragm in a light microscope? [U] / [CO2]

[10x1 = 10]

Part E

Short Answer Type Questions

Answer any **4** Questions

Each Question carries **3** marks

33. Outline the significant contributions of Louis Pasteur [K] / [CO1]

34. Describe the procedure for performing a simple stain [U] / [CO2]

35. How do the wavelength of light, numerical aperture, and the optical properties of lenses contribute to the overall resolution of a microscope? [U] / [CO2]

36. Compare hanging drop method with traditional wet mount [U] / [CO2]

37. Illustrate the structure of bacterial cell. [U] / [CO3]

38. Identify the principles underlying sterilization. [A] / [CO4]

[4x3 =12]

Part F

Long Answer Type Questions

Answer any **1** Question

The Question carries **8** marks

39. Define MBC. List out the steps to check the MBC of a newly synthesized drug. [K] / [CO4]

40. Analyse the intra-cytoplasmic features of a bacterial cell [An] / [CO3]

[1x8 =8]